Organised by

Supported by

22nd Asian Breast Disease Association (ABDA) Annual Scientific Meeting

Strategising Multidisciplinary Management of Breast Diseases

THE ROLE OF BREAST MRI IN INITIAL EVALUATION AND POST- TREATMENT ASSESSMENT OF BREAST MASSES IN DENSE BREASTS

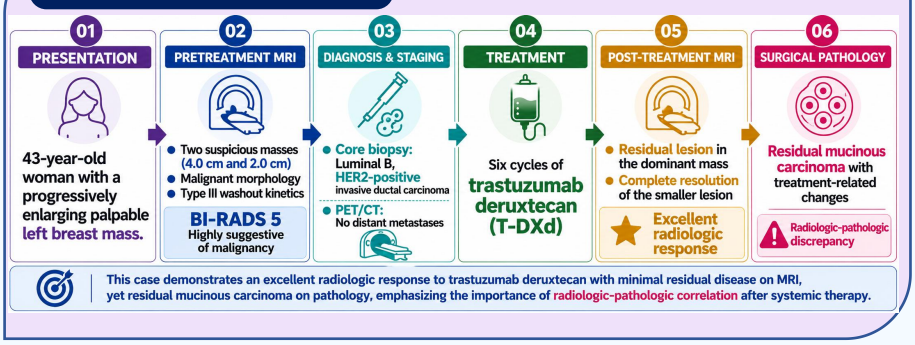
Widya Surya Avanti[1], Lina Choridah[2], Lydia Purna Kuntjoro[3], Sumadi Lukman Anwar[4]

[1]Department of Radiology, RSUD Wates, Kulonprogo, Yogyakarta 55651, Indonesia • [2]Department of Radiology, RSUP Dr Sardjito / Faculty of Medicine, Public Health, and Nursing Universitas Gadjah Mada, Yogyakarta 55281, Indonesia • [3]Department of Radiology, Faculty of Medicine, Universitas Diponegoro, Semarang 50275, Indonesia • [4]Surgical Oncology Division, Department of Surgery, RSUP Dr Sardjito / Faculty of Medicine, Public Health, and Nursing, Universitas Gadjah Mada, Yogyakarta 55281, Indonesia.

BACKGROUND

Breast MRI is highly sensitive for evaluating tumor extent and treatment response in breast cancer. However, an excellent radiologic response may not always correspond to pathological complete response due to tumor heterogeneity and treatment-related changes. This case highlights the role of MRI in pre- and post-treatment assessment of HER2-positive breast cancer and the importance of radiologic–pathologic correlation.

CASE PRESENTATION



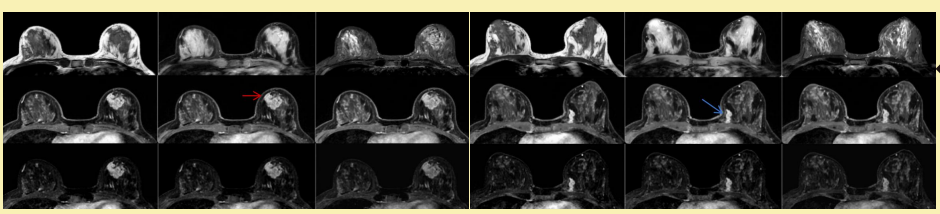


Figure 1. Pretreatment breast MRI. Two suspicious enhancing masses are identified in the left breast (red arrow, larger mass; blue arrow, smaller mass), demonstrating irregular morphology and heterogeneous enhancement. Both lesions show rapid initial enhancement followed by delayed washout (type III kinetic curve), highly suggestive of malignancy (BI-RADS 5)

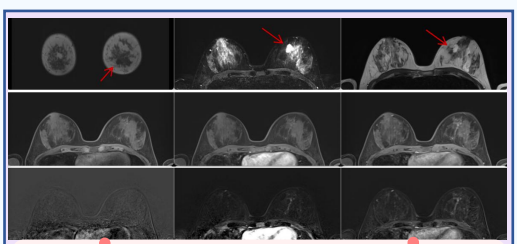


Figure 2. Post-treatment breast MRI. Following trastuzumab deruxtecan therapy, MRI demonstrates a marked treatment response, with only minimal residual enhancement at the site of the previously larger dominant mass (red arrow). The previously identified smaller enhancing mass is no longer visualized, consistent with a complete radiologic response at this site.

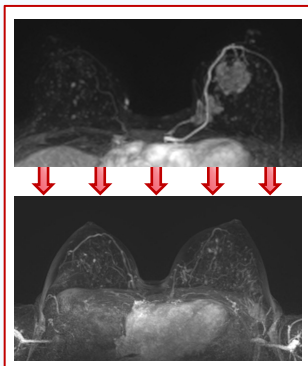


Figure 3. Pre- and post-treatment MIP breast MRI. Pretreatment imaging (top) demonstrates a markedly enhancing left breast mass. Post-treatment imaging (bottom) shows complete resolution of the previously visualized tumor enhancement, consistent with an excellent radiologic response to systemic therapy

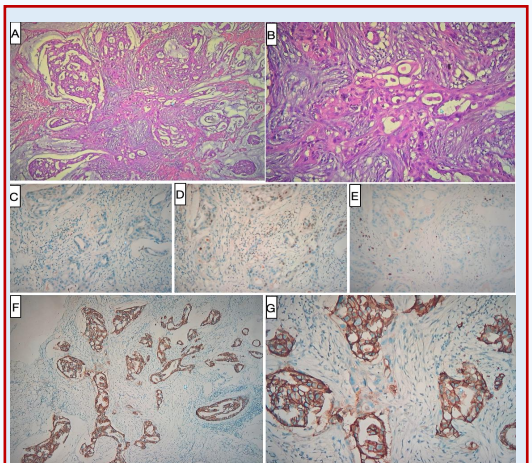


Figure 4. Post-treatment histopathology. H&E sections (A–B) demonstrate residual mucinous carcinoma with invasive tumor cells embedded within abundant extracellular mucin and treatment-altered stroma. Immunohistochemical staining (C–G) demonstrates the biomarker profile of the residual tumor, including strong membranous HER2 expression (F–G). The findings are consistent with residual HER2-positive mucinous carcinoma following systemic therapy.

DISCUSSION

- Breast MRI is valuable for both pretreatment evaluation and post-treatment response assessment. Pretreatment MRI accurately demonstrated multifocal disease, tumor extent, and malignant enhancement characteristics, providing a reliable baseline for treatment planning.
- Following systemic therapy, MRI showed marked tumor regression, indicating an excellent radiologic response. Interestingly, while pretreatment core biopsy revealed invasive ductal carcinoma, the residual tumor at surgery showed mucinous carcinoma with treatment-related changes.
- This histopathologic shift may reflect intratumoral heterogenecity, selective treatment response, or therapy-induced morphologic alterations rather than true transformation of the tumor subtype. These findings highlight that marked radiologic response may not correspond to complete pathologic response and emphasize the complementary roles of MRI and histopathology in post-treatment evaluation.

CONCLUSION / TAKE-HOME MESSAGE

1. Main finding
MRI showed an excellent response despite residual mass

2. Clinical meaning
Radiologic response may not reflect complete pathologic response.

3. Next step
Radiologic–pathologic correlation remains essential after systemic therapy.

REFERENCES

1. DeMartini WB, Strigel RM, Pinker K, Rahbar H, Wang LC. Magnetic resonance imaging. In: Newell MS, Destounis SV, Leung JWT, DeMartini WB, Lee CH, Eby PR, editors. ACR BI-RADS® v2025 Manual. Reston (VA): American College of Radiology; 2025.
2. Janssen LM, den Dekker BM, Gilhuijs KGA, van Diest PJ, van der Wall E, Elias SG. MRI to assess response after neoadjuvant chemotherapy in breast cancer subtypes: a systematic review and meta-analysis. NPJ Breast Cancer. 2022;8:107.
3. Harbeck N, et al. Neoadjuvant trastuzumab deruxtecan alone or followed by paclitaxel, trastuzumab, and pertuzumab for high-risk HER2-positive early breast cancer (DESTINY-Breast11): a randomised, open-label, multicentre, phase III trial. Ann Oncol. 2025. doi:10.1016/j.annonc.2025.10.019.
4. Kuzmiak CM, Calhoun BC. Pure mucinous carcinoma of the breast: radiologic-pathologic correlation. J Breast Imaging. 2023;5(2):180-7. doi:10.1093/jbi/wbac084.
5. Geyer CE Jr, et al. Trastuzumab deruxtecan in residual HER2-positive early breast cancer. N Engl J Med. 2026;394(9):845-57. doi:10.1056/NEJMoa2514661. New England Journal of Medicine

ACKNOWLEDGEMENTS